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the polynomial associated with a message. Supposing that $G(x)$ is an r generator polynomial, the message may be coded by carrying out the following operations:

– multiply $I(x)$ by x^r (or add r zeros at the end of $I(x)$);

– decompose $I(x)x^r$ into the form:

$$\frac{I(x)x^r}{G(x)} = Q(x) + R(x) \qquad [1.26]$$

– determine the cyclic polynomial $T(x)$:

$$T(x) = I(x)x^r - R(x) \qquad [1.27]$$

The polynomial $T(x)$ is a multiple of $G(x)$. It corresponds to a representation of data to which redundant bits have been appended.

Errors are detected by verifying the divisibility of $T(x)$ by $G(x)$.

NOTE 1.5.– Expressions used for the generator polynomials vary by application areas:

– CRC[2]-3-GSM: $G(x) = x^3 + x + 1$;

– CRC-4-ITU: $G(x) = x^4 + x + 1$;

– CRC-8-CCITT: $G(x) = x^8 + x^2 + x + 1$;

– CRC-16-CCITT: $G(x) = x^{16} + x^{12} + x^5 + 1$;

– CRC-32-IEEE: $G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$;

– CRC-64-ISO: $G(x) = x^{64} + x^4 + x^3 + x + 1$.

EXAMPLE 1.28.– Let us consider the initial information 101101, with which the polynomial $I(x) = x^5 + x^3 + x^2 + 1$ can be associated. Using the polynomial generator of the form $G(x) = x^3 + x + 1$ ($r = 3$), the form of the word to be transmitted, or the polynomial $T(x)$, is determined by proceeding as per the steps:

– multiplication of $I(x)$ by x^r gives the product $I(x)x^r = 101101000$;

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– the division of $I(x)x^r$ by $G(x)$ yields the quotient $Q(x)$ = 100001 and the remainder $R(x)$ = 011;

– the polynomial $T(x)$ is finally obtained by appending the r bits of $R(x)$ to the end of $I(x)$, that is: $T(x)$ = 101101011.

In the form $T(x) + E(x)$, the information is assumed to be affected by the error E. With a code based on a polynomial generator $G(x)$, we can detect:

– all the single errors (E = 10 … 0);

– all the double errors (E = 10 … 010 … 0) if $G(x)$ has a factor with at least three terms;

– all the errors relating to an odd number of bits (E has an odd number of bits at 1) if $x + 1$ divides $G(x)$;

– all the series of errors (E = 0 … 01 … 10 … 0) of length smaller than the degree of $R(x)$;

– most of the long series of errors.

1.14. Exercises

EXERCISE 1.1.– Conversions

1) Convert the following numbers to binary:

a) 37_{10} b) 15_{10} c) 187_{10} d) $2\,014_{10}$ e) $2\,016_{10}$ f) 2.75_{10}
g) 25.25_{10} h) 243.3125_{10} i) 0.0625_{10} j) 62_8 k) 277_8 l) 12.6_8
m) 476.35_8 n) 92_{16} o) $37FD_{16}$ p) $7FF_{16}$ q) $1A6_{16}$ r) $2C0_{16}$
s) $1F.C_{16}$ t) $9.F_{16}$ u) $A7.EC_{16}$

2) Convert the following numbers to decimal:

a) 10110_2 b) 10001_2 c) 10001101_2 d) 100100001001_2 e) 1111010111_2
f) 1011.101_2 g) 10011011001.10110_2 h) 30_8 i) 115_8 j) 55.4_8
k) 270.54_8 l) 356_{16} m) $2AF_{16}$ n) $2C1_{16}$ o) $10FF_{16}$
p) $1FCFA_{16}$ q) $DADA.C_{16}$ r) $F.4_{16}$ s) $EBA.C_{16}$

3) Convert the following numbers to hexadecimal:

a) 320_{10} b) $6\,861_{10}$ c) $65\,535_{10}$ d) 100_8 e) 62.4_8 f) 500.25_8
g) 10001101_2 h) 1001000110100011110_2 i) 10000.1_2
j) 1000000.0000111_2 k) 1000111001.01_2

4) Convert the following BCD numbers to decimal:

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a) $0001\ 1000\ 0100_{BCD}$ b) $0100\ 1001\ 0010_{BCD}$
c) $1001\ 0111\ 0101\ 0010_{BCD}$ d) $0111\ 0111\ 0111\ 0101_{BCD}$

5) How many bits are required for the binary representation of the decimal numbers from 0 to 511?

6) What is the largest number that can be represented in 16-bit binary numeration system?

7) a) Determine the binary representation of the decimal number 10.05 with an absolute error equal to 0.005.

b) Represent the decimal number 0.452 in binary numeration system with a relative error of 0.1%.

EXERCISE 1.2.– Representation of numbers and data

1) Let X be an unsigned n-bit integer. Verify that $2^n - X$ represents the two's complement of $-X$.

2) Determine the 8-bit two's complement of the following numbers: -1_{10}, -17_{10}, -128_{10}.

Convert the following numbers to decimal:

$$01111111_{2C}, \qquad 11001110_{2C}, \qquad 10001000_{2C}.$$

3) Determine the two's complement representations of the numbers $-63A_{16}$ and $-8AC_{16}$.

4) Convert the following numbers to ASCII codes:

a) 1_{10}, b) 107_{10}, c) 1000010_2.

5) Use ASCII code to translate the following expressions:

$$X = \cos(.7)$$
$$Y = 256 * X$$
PRINT "X=", X; "Y=", Y

6) a) Represent the following numbers in single-precision IEEE-754 standard: 2.75_{10}, $-417\ 680_{10}$.

b) Find the decimal number corresponding to each of the following single-precision IEEE-754 representations:

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$$0\ 10001010\ 01110111000110000000000_{IEE754}$$
$$00000000\ 00000000000000000000001_{IEEE754}$$

EXERCISE 1.3.– Gray code/binary number conversion

1) Convert the following binary numbers to Gray code:

a) 11011_2 b) 101101_2 c) 11000110_2

2) Convert each of the following Gray code to binary number:

a) 1010_{Gray} b) 00010_{Gray} c) $11000010001_{\text{Gray}}$

EXERCISE 1.4.– Correction code

1) Data are to be transmitted, coded by the CRC method, whose polynomial generator is:

$$G(x) = x^3 + x + 1$$

Determine the bits of the message to be transmitted if the initial message is 101101.

What can we say of the transmission if the received message is 100101 011?

2) Using the CRC coding method, represent the message to be transmitted in the case where the initial information is 1011001 and the polynomial generator is of the form:

$$G(x) = x^4 + x + 1$$

1.15. Solutions

SOLUTION 1.1.– Conversions

1) Conversions to binary representation

a) $37_{10} = 100101_2$

b) $15_{10} = 1111_2$

c) $187_{10} = 10111011_2$

d) $2\ 014_{10} = 11111011110_2$

e) $2\ 016_{10} = 2^{11} - 2^5 = 11111100000_2$

f) $2.75_{10} = 10.11_2$

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g) $25.25_{10} = 11001.01$

h) $243.3125_{10} = 11110011.0101_2$

i) $0.0625_{10} = 0.0001_2$

j) $62_8 = 110010_2$

k) $277_8 = 10111111_2$

l) $12.6_8 = 1010.11_2$

m) $476.35_8 = 100111110.011101_2$

n) $92_{16} = 10010010_2$

o) $37FD_{16} = 11011111111101_2$

p) $7FF_{16} = 11111111111_2$

q) $1A6_{16} = 110100110_2$

r) $2C0_{16} = 1111000000_2$

s) $1F.C_{16} = 11111.11_2$

t) $9.F_{16} = 1001.1111_2$

u) $A7, EC_{16} = 10100111.111011_2$

2) Conversion to decimal representation

a) $10110_2 = 22_{10}$

b) $10001_2 = 17_{10}$

c) $10001101_2 = 141_{10}$

d) $100100001001_2 = 2313_{10}$

e) $1111010111_2 = 983_{10}$

f) $1011.101_2 = 11.625_{10}$

g) $10011011001.10110_2 = 1241.6875_{10}$

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h) $30_8 = 36_{10}$

i) $115_8 = 77_{10}$

j) $55.4_8 = 45.5_{10}$

k) $270.54_8 = 184.6875_{10}$

l) $356_{16} = 854_{10}$

m) $2AF_{16} = 687_{10}$

n) $2C1_{16} = 705_{10}$

o) $10FF_{16} = 4351_{10}$

p) $1FCFA_{16} = 130298_{10}$

q) $DADA.C_{16} = 56026.75_{10}$

r) $F.4_{16} = 15.25_{10}$

s) $EBA.C_{16} = 3770.75_{10}$

3) Conversion to hexadecimal representation

a) $320_{10} = 140_{16}$

b) $6\ 861_{10} = 1ACD_{16}$

c) $65\ 535_{10} = 16^4 - 1 = FFFF_{16}$

d) $100_8 = 40_{16}$

e) $62.4_8 = 32.8_{16}$

f) $500.25_8 = 140.54_{16}$

g) $10001101_2 = 8D_{16}$

h) $1001000110100011110_2 = 48D1E_{16}$

i) $10000.1_2 = 10.8_{16}$

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j) $1000000.0000111_2 = 40.0E_{16}$

k) $1000111001.01_2 = 239.4_{16}$

4) BCD – Decimal conversion

a) $0001\ 1000\ 0100_{BCD} = 184_{10}$

b) $0100\ 1001\ 0010_{BCD} = 492_{10}$

c) $1001\ 0111\ 0101\ 0010_{BCD} = 9\ 752_{10}$

d) $0111\ 0111\ 0111\ 0101_{BCD} = 7\ 775_{10}$

5) How many bits are required for the binary representations of the decimal numbers from 0 to 511?

With *k* bits, only the decimal numbers from 0

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